

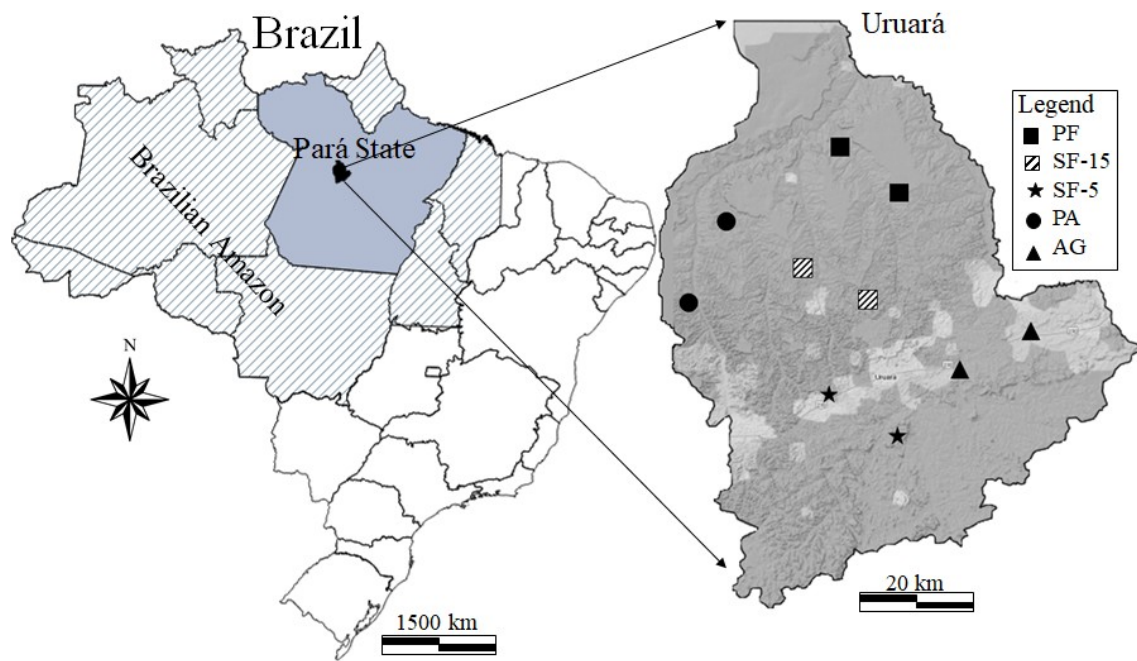


Figure S1. Study region of municipality of Uruará, state Pará, northern Brazil. Location of the sampling sites per ecosystem: PF= Primary Forest; SF-15= Secondary forest with 15 years of regeneration; SF-5= Secondary forest with 5 years of regeneration; AG= Agriculture; PA= Pasture.

Table S1 Generic characteristics of the ecosystems sampled in the municipality of Uruará, state Pará, northern Brazil. Adapted from Darinka et al. 2021; Cajaiba et al. 2018

Ecosystems	N. areas	Characteristics
Primary forest – PF	02	Composed of vegetation whose facies is an upper canopy formed mainly by green trees, with crowns that touch each other, creating a dense and enclosed canopy all year round. In this ecosystem no traces of anthropic action were verified (for example, trails, residues, fires, selective cutting of wood, agricultural activities, among others). Characterized also by a high amount of litter, which contributes to the maintenance and fertility of the soil, and serves as food and shelter for several animal species (Da Silva <i>et al.</i> , 2018)
Secondary forest – SF-15 (vegetation with 15 years of regeneration)	02	A secondary forest (or second-growth forest) is a forest or woodland area that has been re-grown after abandonment of agriculture. Biological diversity gradually increases if there are primary remnants to supply seeds. The average height of the trees is over 12 meters and the average diameter is over 14 centimeters. This type of forest arises from natural succession of vegetation, resulting from abandonment after a period of use, a very common action in the Amazon region (Vieira <i>et al.</i> , 2008), considered an understory, presents dense vegetation, with an intermediate size, greater than 3 meters of height and also with a high amount of litter on the ground (Da Silva <i>et al.</i> , 2018).
Secondary forest –SF-5 (vegetation with five years of regeneration)	02	This type of forest is an embryonic stage of plant succession and consequently presents a simple plant composition, therefore, with little canopy coverage. These environments have high soil temperature and acidity, little accumulated litter, and a greater presence of shrubs (Da Silva, <i>et al.</i> , 2018). The average height of trees is no more than four meters and the diameter of the main trees can reach eight centimeters. This phase typically lasts up to six years and in some cases up to ten years, depending on soil quality and / or seed bank..
Agriculture – AG (cocoa plantations)	02	Agroecosystem represented by areas planted with cacao crops (<i>Theobroma cacao</i> L.) in the form of monocultures. Vegetation can reach 5 to 8m in height, with a closed canopy and a large amount of litter and old trunks. Due to the constant cleaning process, there is little presence of shrubs (Santos <i>et al.</i> , 2015).
Pasture for extensive livestock – PA	02	Forest vegetation consisting predominantly of exotic grasses, namely of the genus <i>Brachiaria</i> spp, used for extensive livestock farming (e.g., raising cattle). This environment is characterized by a high incidence of sunlight, soil compacted by cattle trampling, and absence of litter accumulation in the soil (Falcão <i>et al.</i> , 2015).

Da Silva, WB; Périco, E; Dalzochio, MS; Santos, M; Cajaiba, RL. (2018). Are litterfall and litter decomposition processes indicators of forest regeneration in the neotropics? Insights from a case study in the Brazilian Amazon. *Forest Ecology and Management*, v. 429, p. 189-197.

Falcão, JCF; Dáttilo, W.; Izzo, TJ. (2015). Efficiency of different planted forests in recovering biodiversity and ecological interactions in Brazilian Amazon. *Forest Ecology and Management*, 339, p. 105–111.

Santos, JRM; Mariano, CSF; Martins, LCB; Lacau, LSR; Delabie, JHC. (2015). Assembleias de formigas epigeias (Hymenoptera: Formicidae) em um cacauzal e remanescente de floresta da Amazônia Oriental, Brasil. *Agrotrópica* 27, p. 149 -160.

Vieira, ICG; Toledo, PM; Silva, JMC; Higuchi, H. (2008). Deforestation and threats to the biodiversity of Amazonia. *Brazilian Journal of Biology*, 68, p. 949-956.

Table S2 Specification of the methodology used for monitoring the environmental variables associate with the ecosystems studied.

Variables	Specification	Monitoring Methodology
Temperature – T	Celsius (°C)	Measured during the traps installation and removal with a portable weather station (model Oregon Scientific WMR200A).
Humidity – H	Humidity (%)	Measured during the traps installation and removal with a portable weather station (model Oregon Scientific WMR200A).
Precipitation – P	Precipitation (mm)	Measured during the traps installation and removal with a portable weather station (model Oregon Scientific WMR200A).
Circumference at Breast Height – CBH	Centimeters (cm)	Trunk diameter was taken at breast height (1.3 m) for the trees.
Circumference at Ankle Height - CAH	Centimeters (cm)	The diameter was measured at the ankle height (CAH = 0.1 m) for the shrubs.
Canopy Cover – CC	Percentage (%)	Calculated with a convex spherical densiometer (D) Lemmon and assigned the following classes: 0–5%, 6–25%, 26–50%, 51–75%, 76–95% and 96–100%
Richness of Plants – RP	Number of Plant Species/ m ²	The number of tree species was counted in an area of 100 m ² (10 x 10 m) in the vicinity of each pitfall trap.
Abundance of Plants –AP	Number of Plants/ m ²	The number of tree was counted in an area of 100 m ² (10 x 10 m) in the vicinity of each pitfall trap.
Richness of Shrubs – RS	Number of Species of Shrubs/ m ²	The number of shrubs species was counted in an area of 100 m ² (10 x 10 m) in the vicinity of each pitfall trap.
Abundance of Shrubs – AS	Number of Shrubs/ m ²	The number of shrubs was counted in an area of 100 m ² (10 x 10 m) in the vicinity of each pitfall trap.
Percentage of Exposed Soil	Percentage (%)/ m ²	The percentage exposed soil in each quadrant was estimated in different percentage classes (0-5, 6-25, 26-50, 51-75, 76-95, 96-100%)
percentage of vegetation cover – GC	Percentage (%)/ m ²	The percentage green cover (vegetation up to 1 m height) in each quadrant was estimated in different percentage classes (0-5, 6-25, 26-50, 51-75, 76-95, 96-100%)
percentage of leaf litter cover – LLC	Percentage (%)	The percentage of litter in each quadrant was estimated in different percentage classes (0-5, 6-25, 26-50, 51-75, 76-95, 96-100%).
Height of Leaf Litter – HLL	Centimeters (cm)	Litter height was measured using a ruler at five points inside the square (near each corner and in the center)

Table S3 Description of functional traits used in this study.

Functional traits	Trait specificity	Description
Ecosystem specificity (ES_FT)	Forest specialists (FS)	Species exclusively captured in primary (PF) or secondary forests (SF).
	Open and semi-open environment specialists (OS)	Species exclusively captured in pastures (PA) and cocoa agroforests (AG).
	Forest generalists (FG)	Species captured in forests (PF and SF)
	Ecosystem generalists (EG)	Species captured in all ecosystems
Trophic type (TT_FT)	Cryptic Predators (CP)	Small species that nest primarily in the soil, leaf and rotting logs, where they specialize on living arthropods or their eggs
	Cryptic Omnivores (CO)	Small species nesting within the soil, leaf and rotting logs, where they are generalist predators and scavengers
	Epigaeic Predators (EP)	Medium-to large-sized species that forage on the litter surface and are specialist predators of other arthropods
	Epigaeic Omnivores (EO)	Medium-to large-sized species that forage on the litter surface and are generalist predators and scavengers
	Arboreal Dominants (AD)	Highly active and aggressive species with large colony sizes, nesting in trees
	Arboreal Subordinates (AS)	Other tree-nesting species unspecialized and poorly competitive, ruderal species
	Army Ants (AA)	Highly aggressive, nomadic species with legionary recruitment
	Leaf-cutting Attini (LA)	Highly active and aggressive, polymorphic species with large colony sizes that use leaves to cultivate a symbiotic fungus
	Non leaf-cutting Attini (NA)	Monomorphic, non-aggressive species with small colony sizes that use other organic material to cultivate a symbiotic fungus
	Opportunists (OP)	Species considered omnivorous and

opportunistic feeders, which harvest
plant exudates, scavenge, and capture
live prey as these are encountered.

Table S4 Total number of species (and morphospecies) of ants in the different ecosystems studied in the Brazilian Amazon, and its classification in "Trophic type" and "Ecosystem specificity". PF = primary forest; SF-15 = Secondary forest (15 years of regeneration); SF-5 = Secondary forest (5 years of regeneration); AG = Agriculture (Cocoa); PA = Pasture.

Morphospecies/Species	PF	FS-15	FS-5	AG	PA	Trophic type	Ecosystem specificity
<i>Acromyrmex</i> sp1	1	0	0	0	0	Leaf-cutting Attini	Forest specialists
<i>Acromyrmex</i> sp2	2	1	0	0	0	Leaf-cutting Attini	Forest specialists
<i>Acromyrmex</i> sp3	0	2	0	0	0	Leaf-cutting Attini	Forest specialists
<i>Anochetus</i> sp1	3	0	0	0	0	Epigaeic Predators	Forest specialists
<i>Anochetus</i> sp2	2	0	0	0	0	Epigaeic Predators	Forest specialists
<i>Anochetus</i> sp3	2	0	0	0	0	Epigaeic Predators	Forest specialists
<i>Apterostigma</i> sp1	5	2	0	0	0	Non-Leaf-Cutting Attini	Forest specialists
<i>Apterostigma</i> sp2	2	0	0	0	0	Non-Leaf-Cutting Attini	Forest specialists
<i>Apterostigma</i> sp3	6	16	0	0	0	Non-Leaf-Cutting Attini	Forest specialists
<i>Apterostigma</i> sp4	6	9	0	0	0	Non-Leaf-Cutting Attini	Forest specialists
<i>Apterostigma urichii</i>	9	0	0	0	0	Non-Leaf-Cutting Attini	Forest specialists
<i>Atta cephalotes</i>	0	0	0	0	4	Leaf-cutting Attini	Open specialists
<i>Atta sexdens</i>	3	4	7	0	5	Leaf-cutting Attini	Ecosystem generalists
<i>Atta</i> sp1	17	0	0	6	0	Leaf-cutting Attini	Ecosystem generalists
<i>Atta</i> sp2	21	0	0	0	0	Leaf-cutting Attini	Forest specialists
<i>Azteca alfari</i>	2	4	0	0	0	Arboreal Subordinates	Forest specialists
<i>Azteca</i> sp	108	0	0	0	0	Arboreal Subordinates	Forest specialists
<i>Brachymyrmex brevicornis</i>	0	0	1	1	0	Opportunists	Ecosystem generalists
<i>Brachymyrmex</i> sp1	111	55	11	7	3	Opportunists	Ecosystem generalists
<i>Brachymyrmex</i> sp2	0	31	0	0	0	Opportunists	Forest specialists
<i>Camponotus cingulatus</i>	0	20	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Camponotus melanoticus</i>	0	0	0	3	0	Epigaeic Omnivores	Open specialists
<i>Camponotus sanctaefidei</i>	0	4	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Camponotus senex</i>	6	3	4	2	58	Epigaeic Omnivores	Ecosystem generalists

<i>Camponotus sexguttatus</i>	0	0	0	0	5	Epigaeic Omnivores	Ecosystem generalists
<i>Camponotus</i> sp1	1	7	2	31	0	Epigaeic Omnivores	Ecosystem generalists
<i>Camponotus</i> sp2	19	5	13	82	0	Epigaeic Omnivores	Ecosystem generalists
<i>Camponotus</i> sp3	1	13	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Camponotus</i> sp4	21	1	15	12	4	Epigaeic Omnivores	Ecosystem generalists
<i>Camponotus</i> sp5	388	139	82	23	1	Epigaeic Omnivores	Ecosystem generalists
<i>Camponotus</i> sp6	10	0	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Camponotus substitutus</i>	0	2	0	39	0	Epigaeic Omnivores	Open specialists
<i>Carebara</i> sp	1	0	1	0	0	Cryptic Omnivores	Forest generalists
<i>Carebara urichi</i>	29	4	1	2	1	Cryptic Omnivores	Ecosystem generalists
<i>Cephalotes cordatus</i>	0	1	0	0	0	Arboreal Subordinates	Forest specialists
<i>Cephalotes pusillus</i>	1	0	0	0	3	Arboreal Subordinates	Ecosystem generalists
<i>Cephalotes</i> sp	0	7	0	9	0	Arboreal Subordinates	Ecosystem generalists
<i>Crematogaster</i> sp1	21	16	0	6	0	Arboreal Subordinates	Ecosystem generalists
<i>Crematogaster</i> sp4	22	14	8	7	3	Arboreal Subordinates	Ecosystem generalists
<i>Crematogaster</i> sp6	22	11	21	3	12	Arboreal Subordinates	Ecosystem generalists
<i>Crematogaster limata</i>	99	23	12	9	5	Arboreal Subordinates	Ecosystem generalists
<i>Crematogaster victima</i>	0	0	0	0	3	Arboreal Subordinates	Open specialists
<i>Cyphomyrmex laevigatus</i>	0	0	1	0	1	Non Leaf-cutting Attini	Ecosystem generalists
<i>Cyphomyrmex major</i>	0	0	0	1	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Cyphomyrmex</i> sp	0	0	0	0	17	Non Leaf-cutting Attini	Open specialists
<i>Cyphomyrmex</i> sp1	2	0	0	0	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Dinoponera gigantea</i>	133	58	28	19	3	Epigaeic Predators	Ecosystem generalists
<i>Dinoponera quadriceps</i>	4	0	0	0	0	Epigaeic Predators	Forest specialists
<i>Dolichoderus gagates</i>	3	0	1	0	0	Arboreal Dominants	Forest generalists
<i>Dolichoderus</i> sp1	9	3	7	5	9	Arboreal Dominants	Ecosystem generalists
<i>Dolichoderus</i> sp2	5	2	2	3	11	Arboreal Dominants	Ecosystem generalists
<i>Dolichoderus</i> sp5	2	0	6	2	0	Arboreal Dominants	Ecosystem generalists

<i>Dolichoderus</i> sp6	0	4	0	0	0	Arboreal Dominants	Forest specialists
<i>Dorymyrmex</i> sp	0	0	2	5	0	Opportunists	Ecosystem generalists
<i>Ectatomma brunneum</i>	44	34	19	28	154	Opportunists	Ecosystem generalists
<i>Ectatomma lugens</i>	371	209	0	71	1	Opportunists	Ecosystem generalists
<i>Ectatomma</i> sp	40	19	12	33	1	Opportunists	Ecosystem generalists
<i>Gnamptogenys moelleri</i>	46	21	15	9	54	Epigaeic Omnivores	Ecosystem generalists
<i>Gnamptogenys striatula</i>	38	0	2	0	0	Epigaeic Omnivores	Forest generalists
<i>Gnamptogenys rastrata</i>	0	0	0	1	0	Epigaeic Omnivores	Open specialists
<i>Gnamptogenys</i> sp1	9	5	3	1	0	Epigaeic Omnivores	Ecosystem generalists
<i>Gnamptogenys</i> sp2	4	0	2	0	0	Epigaeic Omnivores	Forest generalists
<i>Leptogenys arcuata</i>	2	0	0	3	0	Epigaeic Predators	Ecosystem generalists
<i>Leptogenys dasygyna</i>	0	0	0	2	0	Epigaeic Predators	Open specialists
<i>Leptogenys</i> sp	0	0	1	1	0	Epigaeic Predators	Ecosystem generalists
<i>Myrmicocrypta foreli</i>	0	11	2	0	0	Non Leaf-Cutting Attini	Forest generalists
<i>Myrmicocrypta</i> sp1	2	8	0	0	0	Non Leaf-Cutting Attini	Forest specialists
<i>Myrmicocrypta</i> sp2	2	1	0	0	1	Non Leaf-Cutting Attini	Ecosystem generalists
<i>Odontomachus bauri</i>	43	12	9	12	11	Epigaeic Predators	Ecosystem generalists
<i>Odontomachus brunneus</i>	16	31	37	19	2	Epigaeic Predators	Ecosystem generalists
<i>Odontomachus haematodus</i>	118	76	11	4	1	Epigaeic Predators	Ecosystem generalists
<i>Odontomachus meinerti</i>	13	31	0	0	0	Epigaeic Predators	Forest specialists
<i>Odontomachus</i> sp	2	0	0	0	1	Epigaeic Predators	Ecosystem generalists
<i>Pachycondyla crassinoda</i>	200	89	16	12	15	Epigaeic Predators	Ecosystem generalists
<i>Pachycondyla harpax</i>	159	311	69	48	52	Epigaeic Predators	Ecosystem generalists
<i>Pachycondyla striata</i>	0	0	2	0	0	Epigaeic Predators	Forest generalists
<i>Pachycondyla</i> sp1	43	0	0	0	0	Epigaeic Predators	Forest specialists
<i>Pachycondyla</i> sp2	9	13	0	0	0	Epigaeic Predators	Forest specialists
<i>Paraponera clavata</i>	4	0	0	0	0	Unclassified	Forest specialists
<i>Paraponera</i> sp	1	0	0	0	0	Unclassified	Forest specialists

<i>Paratrechina longicornis</i>	0	0	3	0	0	Cryptic Omnivores	Forest generalists
<i>Paratrechina</i> sp	9	2	1	0	1	Cryptic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp1	13	39	59	11	7	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp2	113	65	16	41	69	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp3	42	11	27	67	56	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp4	10	29	79	194	107	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp5	15	0	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Pheidole</i> sp6	37	56	37	103	107	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp7	6	0	0	0	0	Epigaeic Omnivores	Forest specialists
<i>Pheidole</i> sp8	0	0	0	54	0	Epigaeic Omnivores	Open specialists
<i>Pheidole</i> sp9	5	0	0	0	39	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp10	8	12	19	21	17	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp11	16	12	9	6	4	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp12	11	3	16	15	10	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp13	77	44	40	95	86	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp14	13	12	38	21	9	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp15	37	18	14	13	8	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp16	41	26	24	16	12	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp17	273	12	67	33	12	Epigaeic Omnivores	Ecosystem generalists
<i>Pheidole</i> sp18	26	49	37	15	13	Epigaeic Omnivores	Ecosystem generalists
<i>Piatythyrea</i> sp1	1	0	0	0	0	Unclassified	Forest specialists
<i>Piatythyrea</i> sp2	1	0	0	0	0	Unclassified	Forest specialists
<i>Prionopelta punctulata</i>	4	0	1	0	0	Cryptic Predators	Forest generalists
<i>Pseudomyrmex</i> sp1	1	17	7	3	12	Arboreal Subordinates	Ecosystem generalists
<i>Pseudomyrmex</i> sp2	16	6	11	33	5	Arboreal Subordinates	Ecosystem generalists
<i>Pseudomyrmex</i> sp5	3	0	3	0	0	Arboreal Subordinates	Forest generalists
<i>Sericomyrmex parvulus</i>	11	7	0	0	0	Non Leaf-cutting Attini	Forest specialists
<i>Sericomyrmex</i> sp1	14	11	0	0	1	Non Leaf-cutting Attini	Ecosystem generalists

<i>Sericomyrmex</i> sp2	1	0	0	1	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Sericomyrmex</i> sp3	6	0	0	1	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Solenopsis</i> sp1	37	62	12	12	16	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp2	15	9	3	45	33	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp3	7	15	27	37	45	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp4	28	18	32	34	13	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp5	18	9	11	19	44	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp6	111	12	5	4	14	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp7	8	16	22	27	31	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp8	18	17	0	0	0	Cryptic Omnivores	Forest specialists
<i>Solenopsis</i> sp9	93	32	23	21	30	Cryptic Omnivores	Ecosystem generalists
<i>Solenopsis</i> sp10	5	3	2	4	29	Cryptic Omnivores	Ecosystem generalists
<i>Strumigenys eggersi</i>	1	2	0	0	0	Cryptic Predators	Forest specialists
<i>Strumigenys louisianae</i>	0	0	3	0	0	Cryptic Predators	Forest generalists
<i>Strumigenys urrhobia</i>	0	1	0	0	0	Cryptic Predators	Forest specialists
<i>Tapinoma</i> sp1	2	0	0	0	1	Unclassified	Ecosystem generalists
<i>Tapinoma</i> sp2	7	3	3	0	1	Unclassified	Ecosystem generalists
<i>Tapinoma</i> sp3	53	16	48	0	0	Unclassified	Forest generalists
<i>Trachymyrmex bugnioni</i>	12	8	0	0	0	Non Leaf-cutting Attini	Forest specialists
<i>Trachymyrmex</i> sp1	4	5	3	1	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Trachymyrmex</i> sp2	5	3	33	0	0	Non Leaf-cutting Attini	Ecosystem generalists
<i>Wasmania</i> sp1	63	13	76	34	26	Epigaeic Omnivores	Ecosystem generalists
<i>Wasmania</i> sp2	81	11	15	13	9	Epigaeic Omnivores	Ecosystem generalists

Table S5. Summary of Generalized Linear Mixed-effects Models (GLMMs) relating the response variables Abundance and Richness of the functional traits Ecosystem specificity and Trophic type and Functional diversity (Functional richness - FRic, Functional evenness - FEve and functional redundancy - FRed) to diverse predictor variables. Codes explanation: standard error (SE) for fixed effects and standard deviation (SD) for Random effects, Z values (Z-value), and associated probability values (P-values). R^2m , marginal R^2 ; R^2c , conditional R^2 . Ecosystem: PF: Primary forest, SF-15: Secondary forest (15 years of regeneration), SF-5: Secondary forest (5 years of regeneration), AG: Agriculture (Cocoa), PA: Pasture.

Table A - Comparisons for Abundance and Richness of "Ecosystem specificity" functional traits					
<i>Abundance</i>					
<i>Forest generalists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects					
$(R^2m = 0.776)$	Intercept	9.002	1.121	2.694	<0.01
	PF	0.671	0.220	1.291	<0.001
	SF-15	0.298	0.021	1.001	<0.01
	SF-5	-	-	-	-
	AG	-	-	-	-
	PA	-	-	-	-
Random effects					
$(R^2c-R^2m = 0.446)$	Sampled sites		0.832		
<i>Forest specialists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects					
$(R^2m = 0.016)$	Intercept	-1.002	1.098	-0.102	>0.05
	PF	0.591	0.014	0.555	>0.05
	SF-15	0.666	0.111	1.001	>0.05
	SF-5	-	-	-	-
	AG	-	-	-	-
	PA	-	-	-	-
Random effects					
$(R^2c-R^2m = 0.002)$	Sampled sites		0.072		
<i>Open and semi-open environment specialists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects					
$(R^2m = 0.017)$	Intercept	1.002	0.331	1.014	>0.05
	PF	-	-	-	-
	SF-15	-	-	-	-
	SF-5	-	-	-	-
	AG	0.096	0.001	1.001	>0.05
	PA	1.001	0.019	0.888	>0.05
Random effects					
$(R^2c-R^2m = 0.139)$	Sampled sites		0.650		
<i>Ecosystem generalists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects					
$(R^2m = 0.141)$	Intercept	1.992	2.008	0.027	>0.05

	PF	0.171	0.001	0.232	>0.05
	SF-15	-1.008	1.120	-0.592	>0.05
	SF-5	0.331	0.071	1.084	>0.05
	AG	0.211	1.001	0.109	>0.05
	PA	0.991	1.012	0.546	>0.05
Random effects ($R^2c-R^2m = 0.104$)	Sampled sites		0.414		
<i>Richness</i>					
<i>Forest generalists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects ($R^2m = 0.224$)	Intercept	3.002	0.121	2.694	<0.05
	PF	1.071	0.220	2.291	<0.05
	SF-15	0.917	0.021	0.044	<0.05
	SF-5	0.100	0.032	0.772	<0.05
	AG	-	-	-	-
	PA	-	-	-	-
Random effects ($R^2c-R^2m = 0.198$)	Sampled sites		0.371		
<i>Forest specialists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects ($R^2m = 0.611$)	Intercept	-12.002	3.098	-4.102	<0.05
	PF	3.671	1.321	2.001	<0.05
	SF-15	1.847	0.712	0.204	<0.01
	SF-5	-	-	-	-
	AG	-	-	-	-
	PA	-	-	-	-
Random effects ($R^2c-R^2m = 0.201$)	Sampled sites		0.504		
<i>Open and semi-open environment specialists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects ($R^2m = 0.198$)	Intercept	1.502	0.331	1.014	>0.05
	PF	-	-	-	-
	SF-15	-	-	-	-
	SF-5	-	-	-	-
	AG	0.096	0.001	2.001	>0.05
	PA	1.001	0.019	0.888	>0.05
Random effects ($R^2c-R^2m = 0.088$)	Sampled sites		0.050		
<i>Ecosystem generalists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z value	P value
Fixed effects ($R^2m = 0.098$)	Intercept	0.781	1.014	0.029	>0.05
	PF	0.987	2.421	0.215	>0.05
	SF-15	-1.018	0.721	-0.571	>0.05

	SF-5	0.239	0.211	0.198	>0.05
	AG	2.211	1.371	0.078	>0.05
	PA	0.491	0.112	0.344	>0.05
Random effects					
($R^2c-R^2m = 0.146$)	Sampled sites		0.316		

Table B - Comparisons for Abundance and Richness of "Trophic type" functional traits.

<i>Abundance</i>					
<i>Cryptic Predators</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
($R^2m = 0.031$)	Intercept	-2.053	1.077	-0.094	>0.05
	PF	-1.970	0.407	-0.395	>0.05
	SF-15	0.340	0.197	0.182	>0.05
	SF-5	1.012	0.331	0.654	>0.05
	AG	-	-	-	-
	PA	-	-	-	-
Random effects					
($R^2c-R^2m = 0.111$)	Sampled sites		0.602		
<i>Cryptic Omnivores</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
($R^2m = 0.103$)	Intercept	-7.053	2.077	-3.004	>0.05
	PF	-3.970	0.407	-1.395	>0.05
	SF-15	1.340	0.197	1.082	>0.05
	SF-5	1.012	0.33	1.654	>0.05
	AG	-1.171	0.557	-0.171	>0.05
	PA	-3.098	0.098	-0.166	>0.05
Random effects					
($R^2c-R^2m = 0.202$)	Sampled sites		0.199		
<i>Epigaeic Omnivores</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
($R^2m = 0.106$)	Intercept	1.221	0.002	0.098	<0.05
	PF	-2.981	0.154	-2.104	<0.05
	SF-15	-2.054	0.098	-3.712	<0.05
	SF-5	0.112	0.166	0.614	<0.05
	AG	1.611	1.009	2.986	<0.05
	PA	5.013	0.717	7.098	<0.05
Random effects					
($R^2c-R^2m = 0.244$)	Sampled sites		0.127		
<i>Epigaeic Predators</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
($R^2m = 0.335$)	Intercept	1.012	0.33	1.654	<0.05
	PF	1.171	0.557	0.171	<0.05
	SF-15	3.098	0.098	0.166	<0.05
	SF-5	-0.771	0.544	-0.303	<0.05

	AG	-3.970	0.407	-1.395	<0.05
	PA	-1.340	0.197	-1.082	<0.05
Random effects ($R^2c-R^2m = 0.251$)	Sampled sites		0.586		
<i>Arboreal Dominants</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.401$)	Intercept	5.310	0.061	2.964	<0.01
	PF	1.891	0.177	5.068	<0.01
	SF-15	2.254	0.033	1.996	<0.001
	SF-5	-3.470	0.003	-4.007	<0.001
	AG	1.679	0.908	2.220	<0.01
	PA	-2.887	1.021	-1.001	<0.01
Random effects ($R^2c-R^2m = 0.199$)	Sampled sites		0.826		
<i>Arboreal Subordinates</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.216$)	Intercept	-4.540	2.002	-3.964	<0.01
	PF	-1.866	0.582	-0.299	<0.01
	SF-15	3.212	1.042	4.016	<0.01
	SF-5	2.047	1.003	1.198	<0.01
	AG	1.001	1.008	1.229	<0.001
	PA	-0.122	0.004	-1.900	<0.001
Random effects ($R^2c-R^2m = 0.231$)	Sampled sites		0.232		
<i>Leaf-cutting Attini</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.432$)	Intercept	-7.945	1.961	-5.571	<0.01
	PF	-3.265	0.570	-0.572	<0.01
	SF-15	5.621	1.021	7.670	<0.01
	SF-5	3.582	0.982	2.288	<0.01
	AG	1.751	0.987	2.347	<0.05
	PA	-0.213	0.039	-3.629	<0.05
Random effects ($R^2c-R^2m = 0.209$)	Sampled sites		0.801		
<i>Non-leaf-cutting Attini</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.510$)	Intercept	5.219	0.944	5.481	<0.01
	PF	1.495	0.170	3.572	<0.01
	SF-15	5.382	0.213	8.190	<0.01
	SF-5	3.582	0.982	2.288	<0.01
	AG	0.281	0.017	1.307	<0.05
	PA	-0.213	0.120	-1.029	<0.05
Random effects					

$(R^2c-R^2m = 0.119)$		Sampled sites	0.692		
<i>Opportunists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
$(R^2m = 0.198)$	Intercept	3.017	1.414	2.409	<0.01
	PF	-0.986	0.070	-4.101	<0.01
	SF-15	1.013	0.201	1.004	<0.01
	SF-5	5.582	1.022	4.018	<0.01
	AG	1.212	1.016	2.614	<0.001
	PA	1.001	2.020	3.041	<0.01
Random effects					
$(R^2c-R^2m = 0.432)$	Sampled sites	0.402			
<i>Richness</i>					
<i>Cryptic Predators</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
$(R^2m = 0.100)$	Intercept	0.153	1.077	1.005	>0.05
	PF	0.133	0.011	0.026	>0.05
	SF-15	0.110	0.097	0.098	>0.05
	SF-5	0.012	0.241	0.049	>0.05
	AG	-	-	-	-
	PA	-	-	-	-
Random effects					
$(R^2c-R^2m = 0.066)$	Sampled sites	0.102			
<i>Cryptic Omnivores</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
$(R^2m = 0.013)$	Intercept	-0.553	1.033	-1.004	<0.05
	PF	-1.091	0.007	-0.395	>0.05
	SF-15	2.340	0.017	0.098	>0.05
	SF-5	0.121	0.27	1.004	>0.05
	AG	-3.340	0.057	-2.171	<0.05
	PA	0.078	1.098	- 2.104	<0.05
Random effects					
$(R^2c-R^2m = 0.311)$	Sampled sites	0.201			
<i>Epigaeic Omnivores</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects					
$(R^2m = 0.396)$	Intercept	3.208	1.902	5.117	<0.01
	PF	3.074	1.045	4.001	<0.01
	SF-15	-1.000	0.107	-1.024	<0.05
	SF-5	2.112	0.704	1.510	<0.05
	AG	1.007	0.630	0.014	<0.05
	PA	0.981	1.009	4.001	<0.05
Random effects					
$(R^2c-R^2m = 0.306)$	Sampled sites	0.609			
<i>Epigaeic Predators</i>					

Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.341$)					
	Intercept	7.039	2.493	2.559	<0.0001
	PF	3.063	1.007	2.143	<0.01
	SF-15	1.501	0.798	0.166	<0.05
	SF-5	2.653	1.004	2.100	<0.01
	AG	-0.900	1.147	-1.091	<0.05
	PA	0.340	0.102	2.028	<0.05
Random effects ($R^2c-R^2m = 0.209$)					
	Sampled sites		0.716		
<i>Arboreal Dominants</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.681$)					
	Intercept	3.472	0.518	2.581	<0.05
	PF	1.186	1.733	2.683	<0.01
	SF-15	2.286	0.372	1.611	<0.001
	SF-5	-2.579	0.167	-1.643	<0.001
	AG	2.101	0.581	1.733	<0.01
	PA	1.801	0.704	0.981	<0.05
Random effects ($R^2c-R^2m = 0.101$)					
	Sampled sites		0.704		
<i>Arboreal Subordinates</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.591$)					
	Intercept	5.997	3.033	7.751	<0.01
	PF	1.737	1.832	1.579	<0.001
	SF-15	2.217	0.123	2.912	<0.05
	SF-5	1.234	1.074	1.507	<0.05
	AG	1.048	0.198	0.401	<0.05
	PA	-1.392	1.014	-0.508	>0.05
Random effects ($R^2c-R^2m = 0.186$)					
	Sampled sites		0.600		
<i>Leaf-cutting Attini</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.600$)					
	Intercept	3.722	1.525	4.501	<0.01
	PF	5.294	1.248	0.997	<0.01
	SF-15	2.442	0.799	1.009	<0.01
	SF-5	2.049	0.702	1.836	<0.01
	AG	-0.836	0.516	-1.171	<0.05
	PA	-0.607	1.030	-0.475	<0.05
Random effects ($R^2c-R^2m = 0.017$)					
	Sampled sites		0.466		
<i>Non-leaf-cutting Attini</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.510$)					
	Intercept	2.433	0.199	0.747	<0.01

	PF	2.021	0.091	1.225	<0.01
	SF-15	1.283	0.718	2.366	<0.01
	SF-5	2.272	1.014	1.364	<0.01
	AG	0.727	0.882	0.602	<0.05
	PA	-0.213	0.120	-1.029	<0.05
Random effects ($R^2c-R^2m = 0.119$)	Sampled sites		0.692		
<i>Opportunists</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.307$)	Intercept	0.903	0.221	2.091	<0.01
	PF	-3.171	1.002	-4.001	<0.001
	SF-15	-2.516	1.414	-1.012	<0.01
	SF-5	0.105	0.019	0.961	<0.01
	AG	0.072	1.001	1.401	<0.05
	PA	1.003	0.711	0.997	<0.05
Random effects ($R^2c-R^2m = 0.211$)	Sampled sites		0.711		
Table C - Comparisons for Functional Diversity for the different ecosystems considered					
<i>Functional richness</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.811$)	Intercept	-14.020	4.044	-10.096	<0.001
	PF	-4.255	1.706	-6.694	<0.0001
	SF-15	2.137	1.007	3.221	<0.01
	SF-5	2.122	0.541	4.110	<0.001
	AG	0.641	1.032	0.981	<0.001
	PA	-1.995	1.881	-5.371	<0.01
Random effects ($R^2c-R^2m = 0.467$)	Sampled sites		0.901		
<i>Functional evenness</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.656$)	Intercept	-8.001	2.201	-3.002	<0.01
	PF	1.379	2.001	1.476	<0.01
	SF-15	3.254	2.011	0.640	<0.01
	SF-5	-0.926	1.058	-0.902	<0.01
	AG	-1.042	0.938	-1.000	<0.01
	PA	3.661	2.004	6.902	<0.001
Random effects ($R^2c-R^2m = 0.328$)	Sampled sites		0.341		
<i>Functional redundancy</i>					
Response variable	Predictor variable	Estimate	SE/SD	Z-value	P-value
Fixed effects ($R^2m = 0.656$)	Intercept	3.774	6.403	9.202	<0.05
	PF	2.679	1.832	5.411	<0.01
	SF-15	-2.313	0.188	-2.481	<0.05

	SF-5	5.547	0.631	4.647	<0.05
	AG	4.247	3.266	5.271	<0.05
	PA	-0.345	1.463	-3.017	<0.01
Random effects ($R^2c-R^2m = 0.186$)	Sampled sites		0.359		

Table S6 PERMANOVA results (based on Bray-Curtis similarity) comparing the composition of Functional groups of ants between ecosystems (ES_FT: Permanova F = 15.09, P <0.001; TT_FT: Permanova F = 23.16, P <0.001). The values highlighted in gray correspond to the significance test (p-value). PF= Primary Forest; SF-15= Secondary forest with 15 years of regeneration; SF-5= Secondary forest with 5 years of regeneration; AG= Agriculture; PA= Pasture.

ES_FT: Ecosystem specificity					
	PF	SF-15	SF-5	AG	PA
PF	-	0.001	0.0001	0.0001	0.0001
SF-15	13.44	-	0.001	0.001	0.001
SF-5	19.67	11.13	-	0.05	0.01
AG	35.23	9.13	7.21	-	0.05
PA	33.52	23.11	8.98	8.15	-
TT_FT: Trophic type					
	PF	SF-15	SF-5	AG	PA
PF	-	0.001	0.0001	0.0001	0.0001
SF-15	11.79	-	0.001	0.001	0.001
SF-5	13.23	9.13	-	0.05	0.01
AG	24.17	11.23	6.12	-	0.05
PA	21.88	21.33	7.21	5.15	-